

ASTA 607-COMPUTATIONAL STATISTICS 1

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PRACTICAL EXAMINATION

QUESTION 1

a) Data description

?crabs

#The crabs data frame has 200 rows and 8 columns, describing 5

#morphological measurements on 50 crabs each of two color forms and both sexes,

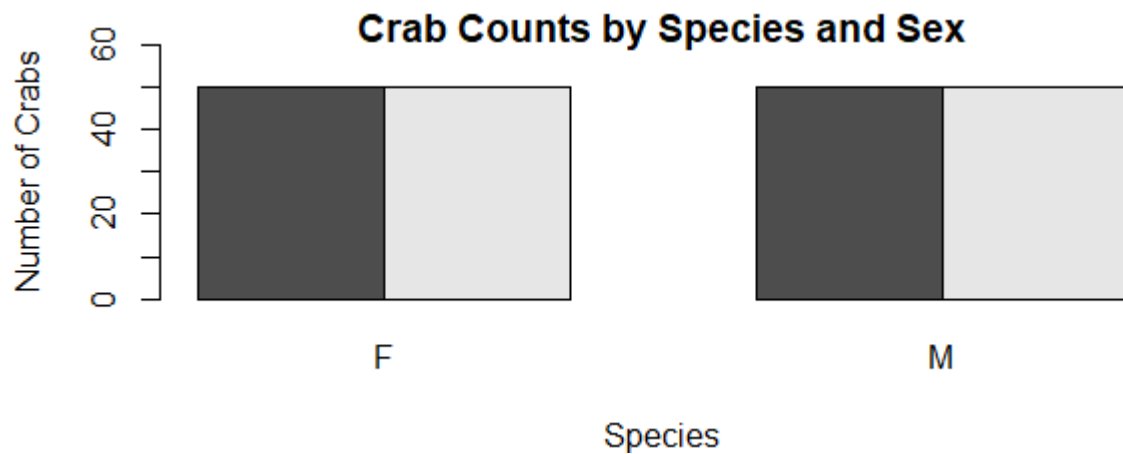
#of the species *Leptograpsus variegatus* collected at Fremantle, W. Australia.

b) Loading data set AND DISPLAY THE STRUCTURE

c) frequency table for species and sex

d)

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e) Comment

There are 50 observations in each combination of species (B, O) and sex (M, F),

#giving 4 groups with equal sample sizes.

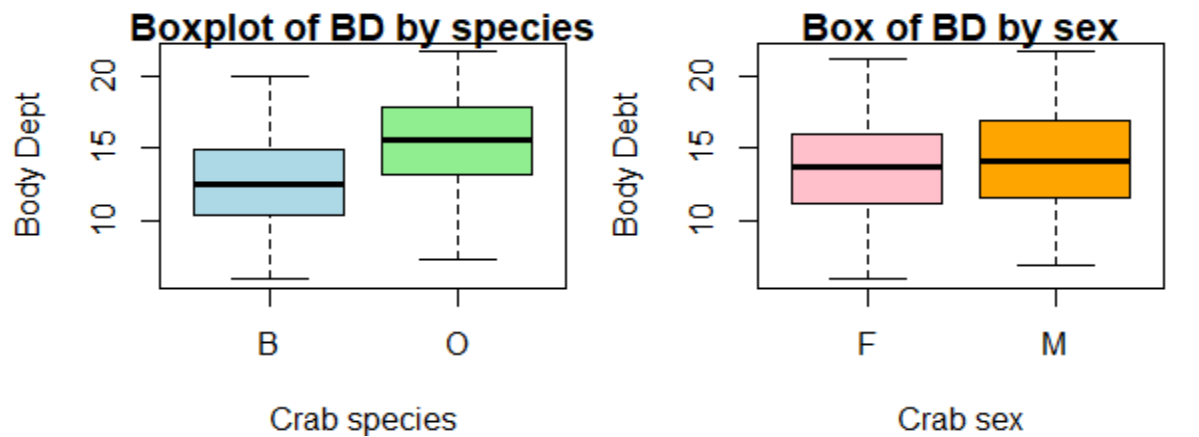
#This means comparisons between groups are not affected by unequal sample sizes.

QUESTION 2

a)



OUTPUT



b) There are no outliers here in this groups.

Central Tendency (Median)

By Species:

Species O has a higher median body depth than Species B.

This suggests Orange crabs tend to have deeper bodies on average.

By Sex:

Males and females have similar medians, though males appear slightly higher.

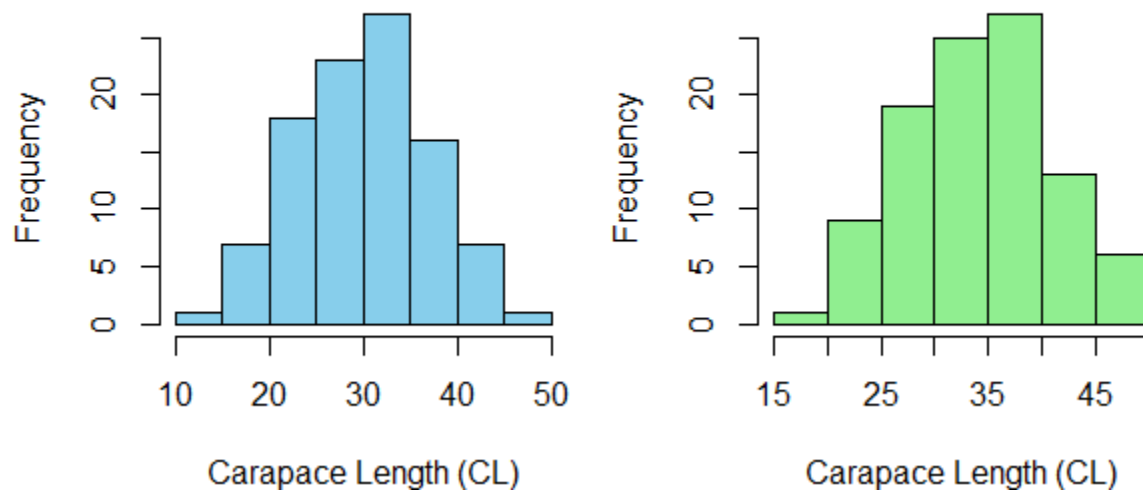
The difference by sex is smaller than the difference by species.

Question 3

a)



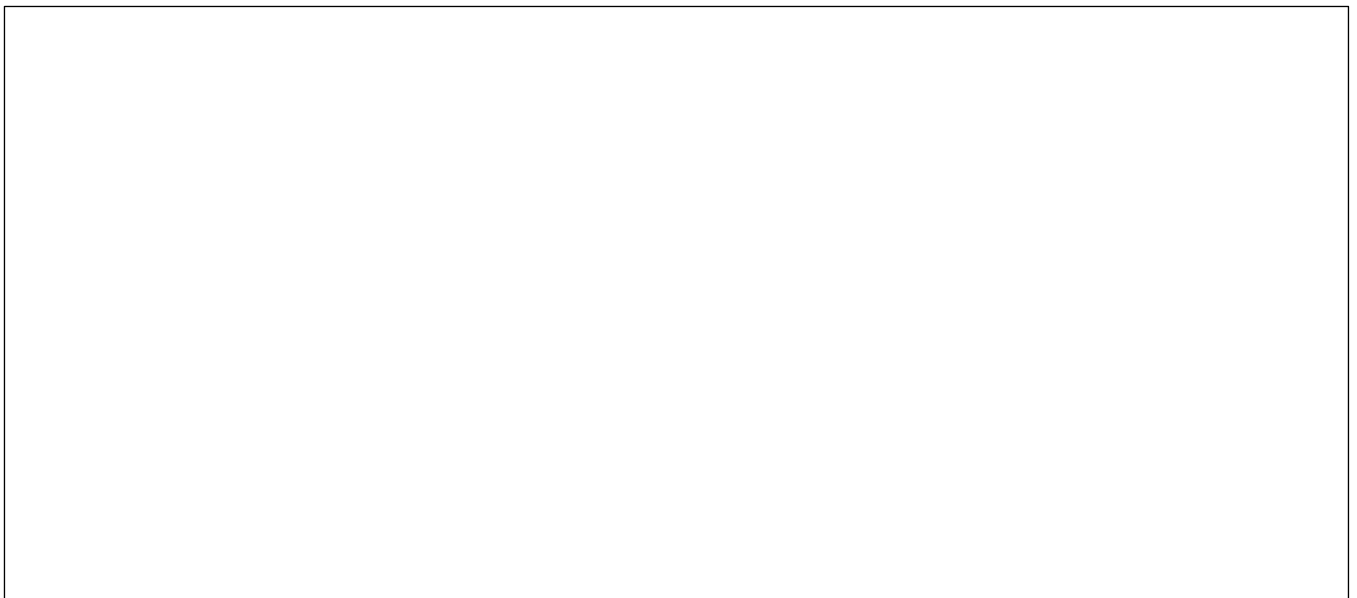
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b) Both species show roughly bell-shaped distributions, indicating that carapace length is approximately normally distributed in each group. The distributions are unimodal, with a single clear peak around the central values, suggesting most crabs have lengths near the mean. Species O appears slightly right-skewed with more larger values, while Species B is more symmetric, indicating slightly less variability toward higher carapace lengths.

QUESTION 4

a)

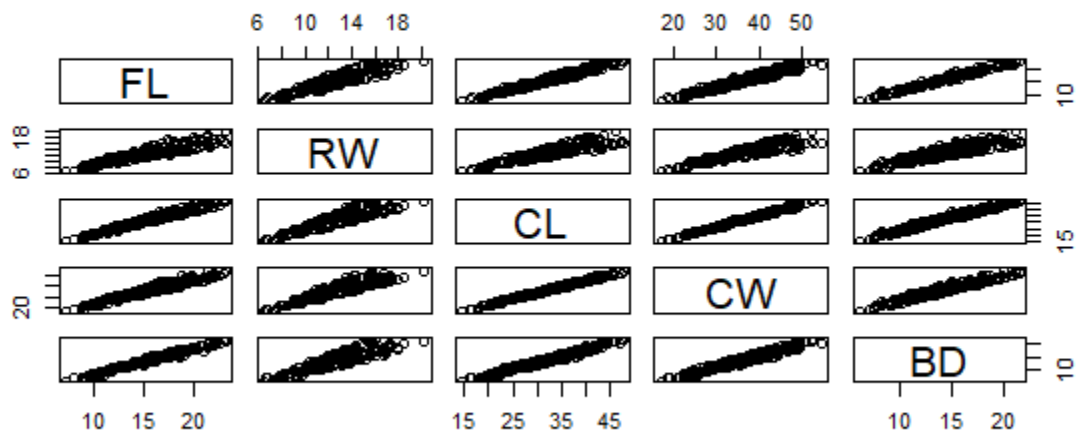


QUESTION 5

a)

OUTPUT:

Scatter plot matrix OF MOPHORLOGICAL MEASUREMENT



b) The diagram above indicated that there exist a positive relationship between the variables